84-375 / 84-675 Geopolitics of Innovation

Carnegie Mellon Institute for Strategy and Technology

Class Location: WEH 4708
Class Times: Monday & Wednesday 2:00 - 3:20pm
Professor: Justin K. Canfil <jcanfil@cmu.edu>

Office Location: Posner 382B
Office Hours: Wednesday, 11:00-1:00pm (sign-up slots)
Syllabus Version: 1.8 (Aug 31 2026)

Course Description: Technology is applied knowledge, and knowledge is power. This course explores how national power is derived from scientific and technological advancements, impacting the global geopolitical landscape. Students will be introduced to interdisciplinary theories about why technological competition occurs, when pivotal shifts happen, why certain policies prevail, and how outcomes are shaped. We will examine what drives innovation in different countries, how nations cooperate or compete in scientific and technological fields, and the role of creativity. We will analyze how competition in research and innovation has evolved over medium- and long-term historical periods, from antiquity to present-day strategic investments. An animating question of the course is the extent to which policymakers can, do, or should influence technological change---vital questions for this century.

Course Objectives: By the end of the semester, students should be able to:

- Understand the state of scholarly debate about what makes nations innovative and the ways in which harnessed innovation contributes to national power;
- Consider how scholarly and policy-oriented thinking about tech competition has developed, situating their own perspectives within these evolving intellectual traditions;
- Understand present-day technology competition in historical context;
- Participate thoughtfully in current political debates about the national research and innovation enterprise.

Prerequisites: No specific prerequisites. Students must have taken at least some prior International Relations coursework and should have general awareness of current events. The course assumes a basic working knowledge of international history, especially in the 20th century. For a refresher, I recommend *J. W. Young and J. Kent, International Relations since 1945: A Global History. Oxford ; New York: Oxford University Press, Feb. 26, 2004, 768 pp., isbn: 978-0-19-878164-6.*

Required Readings: Without exception, weekly readings should be done ***by Monday before the start of class.*** All readings except the following are freely available on Canvas or via the CMU Library. Readings are selected to prompt discussion and debate, and are often paired precisely because the authors disagree. **You are not required to agree with what you read.**

What is expected is that you understand what each author claims, why they claim it, and how the readings interact. Come to class prepared to defend, contest, or extend what you have read.

Questions About the Class: You are most welcome to ask substantive questions about the class during office hours. For answers to questions about assignments, due dates, readings, grading policies, excused absences, and other policies, always consult the syllabus FIRST before reaching out. I recommend contacting me with any questions by email (jcanfil@cmu.edu) as I do not always get Canvas inbox notifications. Please be patient. I will do my best to get back to you within 24-48 hours. I will not reply over the weekend except in exceptional circumstances and I do not expect you to, either.

Grading Scale: Final course grades will be assigned according to the following grading scale:
A (90-100), B (80-89), C (70-79), D (60-69), and R (failing, < 60)

Important Deadlines:

Deliverable	**Due Date**	**Points**	**Percent of Grade**
Participation	–	50	10
Reading Quizzes (up to 3x)*	Unannounced	25 each	5 each
GenAI Exercise	9/9	25	5
Technology Map (Group Presentation)	9/28 - 10/5	50	10
Discussant Duties	9/28 - 10/5	25	5
Midterm	10/7	75	15
Simulation (Strategy Brief)	11/12	50	10
Simulation (Report)	11/30	50	10
Final Exam	12/2	100	20

* Makeup quizzes are not permitted, but you may drop your lowest quiz score. If fewer than 3 quizzes are given, remaining points will default to participation denominator.

- **Participation:** To earn full credit, you must regularly attend class and participate thoughtfully by asking and/or answering questions. This includes having done the readings in advance so that you are prepared to participate in an informed discussion.
- **Reading Quizzes:** Quizzes will consist of 1-2 questions drawn directly from the weekly readings. These are closed-note.
- **GenAI Exercise:** Two components. First, complete the in-class technology definitions worksheet. Second, use our discussions in class to come up with a proposed working

definition for technology. Submit your definition to https://forms.gle/H3BVV3JZYSZni3pq9 by the deadline. You are encouraged (but not required) to use Generative AI to assist you in coming up with a definition. Students who use GenAI are expected to debate it until it produces with a definition you think is satisfactory. **If you do not use GenAI,** please (1) state your definition at the top, then (2) explain alternatives you considered and why you think this is the best candidate formulation. **If you do use GenAI,** please include your initial prompt, an excerpt of the model's initial reply, where you think its definition fell short, and a detailed explanation of how exactly you had to push it to develop and refine a better definition.

- **Technology Map Presentation**: As a team, present for 20 minutes leaving 5-10 minutes for Q&A. Your team should select a "critical technology" area from the NSTC (2024) Report by **9/22**. Choose one example product within that area (an actual or conceptual design). Create one slide for each of the following questions:
    - **(1) What is the product? How does it work?** Discuss core system functions, critical components/subsystems, and architecture in which the components are embedded.
    - **(2) Lifecycle:** Draw a flow diagram to depict the development and diffusion lifecycle. Key milestones: conceptualization → prototype → scaling → dominant design/platform.
    - **(3) Inspiration:** How did the idea originate? Was it potentially inspired by a science fiction concept, military or market demand, both, or something else?
    - **(4) Design:** Is the product you chose recombinant or radical? Why? Include a basic genealogy if preceded by an analogous technology and explain key differences in hardware and use-cases.
    - **(5) Cooperation:** How are development and parts sourced? Which countries are involved at which points? Applicable export control regimes or standards?
    - **(6) Applications:** Spin-on / spin-off / dual-use / single-use / general purpose? Show evidence for your classification and any trajectory shifts (e.g., civilian → military).
    - **(7) Implications:** Is the product you chose an incremental or disruptive innovation? Did it create new markets? Displace dominant companies? Other effects?
    - You should **submit slides and notes to Canvas 24 hours before** your scheduled presentation.
- **Discussant Duties**: Open Q&A by asking the presenters at least one question. Your questions should be derived directly from their presentation material and should be explicitly tied with readings or lecture concepts from Weeks 1-5.
- **Midterm**: Exam will consist of 2 or 3 essay-length questions to be completed during class time. Any material from readings or lectures prior to date is fair game. Open note (handwritten). You may not use a laptop or electronic device.
- **Strategy Brief**: As a team, compile a 1-page assessment of your country's National Innovation System (NIS) using the CIA Factbook: https://www.cia.gov/the-world-factbook/. Assessment should analyze strengths and weaknesses vis-a-vis the other countries in the game for each of the relevant

dimensions. Teams consist of czars (advisors) and at least one executive decisionmaker. Teams may establish any voting rule they choose (eg. unanimity, majority vote, dictator) but must declare this in the brief. Independently, each czar should write 2-3 paragraphs recommending a strategy for their respective area, including a justification for why they think that strategy is optimal. Teams discuss each recommendation internally. Executives should then write 2-3 paragraphs discussing choice of overall strategy (based on chosen voting rule) and whether there was team consensus on every recommendation. Executive summary will form the basis of your Round 1 strategy.

- **Simulation Report**: Based on the Statistics Czar's notes and your own recollection of internal team debates, write 2-3 pages explaining (1) how gameplay unfolded at each stage; (2) team members' beliefs about their country's progress relative to other countries'; and (3) whether, how, and why your Round 1 strategy evolved between stages. (4) If your team "won," explain why, taking into account other teams' potential missteps. If your team did not "win," consider what decisions you could have made differently in retrospect and how they could have changed the result. Teams should divide the labor equally. *(Optionally, please feel free to conclude with any remarks commenting on game mechanics, including what you think should be changed. Feedback will help improve it for future classes!)*
- **Final Exam**: Exam will consist of 2 or 3 essay-length questions to be completed during class time. Any material from readings or lectures is fair game. Open note (handwritten). You may not use a laptop or electronic device.

Extra Credit: Students may earn extra credit through verified attendance at a CMIST event during the semester. Attendance at each event, for up to three events, will add one percent to students' final grades, for a maximum of three percent. A schedule of events can be found at https://www.cmu.edu/ips/news-and-events/index.html. To receive credit, students must attend the talk and then post a one-paragraph reaction here (https://forms.gle/nTy5ruDuQ9idGUY48). In your reaction, you should explain what the one main takeaway or thing you learned from the event was. If you asked a question during the event, you may write about what motivated the question as well as whether you were satisfied with the answer you received. I may ask those who attend to share what they learned in class.

Grade Complaints will not be entertained until 24 hours after the exam/assignment is returned to you or more than one week after the exam/assignment has been returned. If you have a question concerning the grade you have received:

- Wait 24 hours, but no more than 1 week.
- Write an explanation, including the reasons why you think your grade should be changed. Such appeals should refer to specific things in the exam/assignment, and not to vague reasons like "I worked really hard."
- Submit your written complaint/appeal to the instructor, and make an appointment to meet either during office hours.

The second grade will become final on the assignment. It may be higher, lower, or the same.

Policy on Late Work: One full letter grade will be deducted for each day late. Assignments over five days late will not be graded. Documentation is required for serious extenuating circumstances. The instructor should be alerted that an assignment will be late *in advance* of the due date, or as soon as possible in the event of a legitimate emergency.

Regular Attendance: Regular in-person attendance is essential and expected from all students. Absences for genuine illnesses are permitted—do not try to come to class if you are too sick or might get others sick. However, the class moves quickly. If health issues cause you more than two classes, you should make arrangements with me in advance to avoid falling behind. Missing class is not by itself an excuse for unsubmitted work.

Use of Electronic Devices. Use of handheld devices such as cellphones is never permitted. In the past, I have found that laptops and tablets are a distraction for both their users and peers around them. Research on learning has shown that electronic devices in the classroom reduce attention and impair note-taking quality**.** Multitasking is a myth; it is simply divided attention. As a result, **I have reluctantly decided NOT to allow electronic devices during class**, **even for notetaking purposes,** unless required for an assignment**.** As usual, exceptions will be made for students who notify me in advance of approved disability accommodations (see below).

This means you are expected to take handwritten notes. This has multiple advantages. Handwriting forces you to select only the most important pieces of information, instead of dictating the professor's speech verbatim. Then, optionally, I recommend typing up and organizing those notes outside of class. Even if you must use handwritten notes on the exams, this exercise is proven to improve your memory and understanding of the material. Although you are free to study together, you **must take your own notes** and cannot use another student's notes on the exams.

No Recordings or Photographs of Any Kind: *No student may record any classroom activity without express written consent from me. This includes photographs of the slides or of anyone in the class. If you take photos or are found to be recording without permission, you will be immediately asked to leave. Participation grade penalties may apply.* If you have (or think you may have) a disability that requires you to record (eg. because you cannot handwrite your notes), you should contact the Office of Disability Resources to request a accommodation.

Disability Accommodations: Students with documented disability will be accommodated. Students are responsible for registering with Carnegie Mellon Disability Resources. By university policy, accommodations are non-retroactive. If you have a disability and have an accommodations letter from the Disability Resources office, discuss your accommodations and needs with the instructor as early in the semester as possible to ensure that accommodations are provided as appropriate. If you suspect that you may have a disability and would benefit from accommodations but are not yet registered with the Office of Disability Resources, you may contact them at access@andrew.cmu.edu.

Academic Integrity: Students are expected to maintain the highest standards of academic integrity. Plagiarism, cheating, or any other forms of academic dishonesty will not be tolerated and will be subject to disciplinary action consistent with CMU policies. For more information, please consult the CMU's academic integrity policy. Lack of knowledge about the policy on academic honesty is not a reasonable explanation for a violation. If you feel uncertain, reach out to the instructor for clarification.

Policy on Generative AI: Generative AI (GenAI) refers to any technology that can autonomously produce text, images, code, audio, or other content based on prompts or inputs, including but not limited to tools like ChatGPT, DALL·E, GitHub Copilot, and similar systems. Because of inherent reliability problems, GenAI is best used as a starting point or a synthesizer of your own ideas. It should never be used as a sole source of evidence. Always treat any "factual" GenAI output as suspicious until proven otherwise. In this course, students may use GenAI as a study aid (eg. generating practice questions), but be aware that its guidance may be inaccurate. Students who use GenAI in their writing must include a citation for each instance in which they used it, including precisely how it was used, plus the prompt language. If substantive GenAI output is ever copied verbatim, in addition to being accompanied by a citation, it must be included in quotation marks AND italics to distinguish it from human writing. Ideal use of GenAI would include an appendix indicating prompt language/log plus a factual critique of the output. The GenAI policy may be subject to change at any time during the semester.

Student Wellness: All of us benefit from support during times of struggle. You are not alone. There are many helpful resources available on campus and an important part of the college experience is learning how to ask for help. Asking for support sooner rather than later is often helpful. CMU services are available, and treatment does work. You can learn more about confidential mental health services available on campus at: http://www.cmu.edu/counseling/.

Academic Support: For additional academic services which would assist in completing course assignments, including coaching, peer tutoring, communication (eg. writing) and language support, visit the Student Academic Success Center website at https://www.cmu.edu/student-success/. Support is always available (24/7) from Counseling and Psychological Services: 412-268-2922. Consider reaching out to a friend, faculty or family member you trust for assistance getting connected to the support that can help.

Religious Observance: Students are generally excused from class to attend religious observance. However, this is a process that each student must initiate to obtain an explicit permission. Please email me at least 10 days in advance. Failure to do so will result in an unexcused absence.

Respect, Tolerance, and Kindness: Students from all backgrounds and perspectives are welcome in this course. The classroom is a professional setting. Treat others with respect, tolerance, and open-mindeness at all times. Debate is encouraged, as long as disagreements are informed, civil, and in good faith. If you have a preferred nickname and/or pronoun, please

kindly indicate this to me before the first day of class in order to reduce the likelihood that I make a mistake!

Changes to the Syllabus: I reserve the right to amend the syllabus at any time. Any changes will be announced. Students are responsible for adhering to the syllabus. Please let me know if you believe you have detected a mistake.

How to Succeed in This Course: As the instructor of this course, my goal is for everyone in it to succeed. In the meantime, success is also a function of the effort you put in. You should be active, engaged and committed in your learning. Here are some tips:

- Do the reading. The classes are no substitute for the readings. Assignments will demand familiarity with material not covered in class but found in the readings.
- Read critically. Focus on the big picture to glean the main arguments in the texts. Think about the underlying logic of each argument and draw linkages and contrasts among texts. Don't assume the authors are correct—look at the evidence.
- Come to class. Classes will help you contextualize, interpret, and even challenge the readings. They will also include material not in the readings and live discussions.
- Ask questions. Given that this is a discussion-based class, you are encouraged—and expected—to ask questions or raise issues during class. Your comments and questions will not only deepen your own understanding of the subject matter, but will also help the rest of us see new angles.
- Follow relevant issues and events outside of class. The following sources, for example, offer nuanced analyses of current international affairs:

  – *War on the Rocks*
  – *Lawfare Blog*
  – *The National Interest*
  – *Foreign Affairs Magazine*
  – *Foreign Policy Magazine*
  – *The Economist*
  – *Bulletin of the Atomic Scientists*
  – *The Diplomat*

Week 1: Introduction

Required Reading	Length
Kania, Elsa B. 2025. "Science Securitized, Technology Transformed: Innovation and Exploitation in Non-Military Warfare." In *Non-Military Warfare*, Routledge.	27 pages
Zegart, Amy. 2024. "The Crumbling Foundations of American Strength." *Foreign Affairs* 103(5). https://www.foreignaffairs.com/united-states/crumbling-foundations-american-strength-amy-zegart	~15 pages
Weiss, Charles. 2005. "Science, Technology and International Relations." *Technology in Society* 27(3): 295–313. doi:10.1016/j.techsoc.2005.04.004.	18 pages

August 24: Orientation

Lecture:	Syllabus overview Animating themes

August 26: Defining Tech 1

Lecture:	Definitions & concepts

Week 2: Pinning Down Definitions

Required Reading	Length
Cronin, Audrey. 2020. "Technology and Strategic Surprise: Adapting to an Era of Open Innovation." *The US Army War College Quarterly: Parameters* 50(3). doi:10.55540/0031-1723.2675.	14 pages
Thierer, Adam. 2024. "Defining 'Technology.'" *Medium*. https://medium.com/@AdamThierer/defining-technology-0c5c95c8360d (August 19, 2025).	~3 pages

"Xi Jinping: 'Strive to Become the World's Primary Center for Science and High Ground for Innovation.'" 2021. *DigiChina*. https://digichina.stanford.edu/work/xi-jinping-strive-to-become-the-worlds-primary-center-for-science-and-high-ground-for-innovation/	~5 pages

August 31: Defining Tech 2

Lecture:	Definitions & concepts (cont'd)

September 2: NO CLASS (APSA)

Lecture:	Makeup TBD

Week 3: Pinning Down Definitions (cont'd)

Required Reading	Length
Recommend getting ahead on Week 4 readings (90+ pages)	

September 7: NO CLASS

Lecture:	University closed

September 9: Defining Tech 3

Lecture:	Definitions & concepts (cont'd)

Week 4: Inside the Black Box

Required Reading	Length
Summary of Kranzberg's Laws. https://androsaba.github.io/blog/2024/kranzbergs_laws/	~1 page
Godin, Benoît. 2006. "The Linear Model of Innovation: The Historical Construction of an Analytical Framework." *Science, Technology, & Human Values* 31(6): 639–67. doi:10.1177/0162243906291865.	14 pages

Dafoe, Allan. 2015. "On Technological Determinism: A Typology, Scope Conditions, and a Mechanism." ***Science, Technology, & Human Values*** **40(6): 1047–76. doi:10.1177/0162243915579283.**	29 pages
Wyatt, S. 2008. "Ch. 7: Technological Determinism Is Dead; Long Live Technological Determinism." In ***The Handbook of Science & Technology Studies*****, eds. E. J. Hackett, O. Amsterdamska, M. Lynch, and J. Wajcman. Cambridge, MA, USA: MIT Press, 165–81. https://www.dhi.ac.uk/san/waysofbeing/data/data-crone-wyatt-2007b.pdf#page=181.**	16 pages
Kline, Ronald, and Trevor Pinch. 1996. "Users as Agents of Technological Change: The Social Construction of the Automobile in the Rural United States." ***Technology and Culture*** **37(4): 763–95. doi:10.2307/3107097.**	32 pages

September 14: The Genesis Question

Lecture:	Futurology Eureka moments Newtonian hypothesis Ortega hypothesis Sources of inspiration

September 16: The Tech Lifecycle

Lecture:	*Presentation slot signups* Technology lifecycles Linear & nonlinear innovation Tech Determinism Tech Constructivism

Week 5: Knowledge Production & Knowledge Capture

Required Reading	**Length**
"Letter from Albert Einstein & Leo Szilard to FDR." https://thebulletin.org/virtual-tour/albert-einstein-leo-szilard-and-the-letter-that-led-to-the-manhattan-project/ (August 1939).	2 pages
Bush, Vannevar. 1945. ***Letter of Transmittal and Summary of the Report***	Preface, Chapters 1,

"Science: The Endless Frontier."* A Report to the President.** https://www.pi.infn.it/~giorgio/INFN/3M/SciencetheEndlessFrontier.pdf**.** **Kreps, Sarah. 2025. "An Attack on America's Universities Is an Attack on American Power."** ***Foreign Affairs**.** https://www.foreignaffairs.com/united-states/attack-americas-universities-attack-american-power **(July 23, 2025).** **Gaulé, Patrick, and Mario Piacentini. 2013. "Chinese Graduate Students and U.S. Scientific Productivity."** ***Review of Economics and Statistics*** **95(2).**	4, & 5 ~4 pages 3 pages

September 21: Twin Pillars

Lecture:	Technology since antiquity People & Institutions Public goods problem Modern university system The Manhattan Project Operation Paperclip Endless Frontier Talent flows

September 23: Superadditivity

Lecture:	Open Science (networks, proximity, diversity) Appropriability Problem IP vs. trade secrets Valleys of death Delphic agencies NIS & Multihelixes *In-class presentation.*

Week 6: Presentations

September 28: Presentations

Activity:	In-class presentations.

September 30: Presentations

Activity:	In-class presentations.

Week 7: Midterm Week

October 5: Midterm Exam

Activity:	In-class exam.

October 7: NO CLASS

Activity:	Makeup TBD

Week 8: FALL BREAK

No class. Midterm grades: October 21.

Week 9: Tech Competition

Required Reading	Length
Bower, Joseph L., and Clayton M. Christensen. "Disruptive Technologies: Catching the Wave." *Harvard Business Review*. https://hbr.org/1995/01/disruptive-technologies-catching-the-wave (August 14, 2025).	12 pages
Usselman, Steven W. 1992. "From Novelty to Utility: George Westinghouse and the Business of Innovation during the Age of Edison." *Business History Review* 66(2): 251–304. doi:10.2307/3116939.	53 pages (skim)
"Elisha Gray and Alexander Bell Telephone Controversy." 2025. *Wikipedia*. https://en.wikipedia.org/wiki/Elisha_Gray_and_Alexander_Bell_telephone_controversy (October 6, 2025).	12 pages

October 19: Patent Races

Activity:	Screening of *The Current War (2017)*

October 21: Patent Races

Activity:	Screening of *The Current War (2017)*

Week 10: Shortcuts

Required Reading	Length
Monday:	
Gilli, Andrea, and Mauro Gilli. 2019. "Why China Has Not Caught Up Yet:	48 pages

Military-Technological Superiority and the Limits of Imitation, Reverse Engineering, and Cyber Espionage." *International Security* 43(3): 141–89. doi:10.1162/isec_a_00337.	
Calcara, Antonio, Andrea Gilli, Mauro Gilli, Raffaele Marchetti, and Ivan Zaccagnini. 2022. "Why Drones Have Not Revolutionized War: The Enduring Hider-Finder Competition in Air Warfare." *International Security* 46(4): 130–71. doi:10.1162/isec_a_00431.	59 pages
***Weds*:**	
"The Threat Posed by the Chinese Government and the Chinese Communist Party to the Economic and National Security of the United States." *Federal Bureau of Investigation*. https://www.fbi.gov/news/speeches-and-testimony/the-threat-posed-by-the-chinese-government-and-the-chinese-communist-party-to-the-economic-and-national-security-of-the-united-states (July 15, 2025).	1 page
***Letter from US Select Committee on the Chinese Communist Party*. 2025. US House of Representatives. https://selectcommitteeontheccp.house.gov/sites/evo-subsites/selectcommitteeontheccp.house.gov/files/evo-media-document/3.18.2025%20-%20Letter%20to%20Carnegie%20Mellon%20-%20SCC.pdf**	4 pages
Cheung, Tai Ming. 2016. "Innovation in China's Defense Technology Base: Foreign Technology and Military Capabilities." *Journal of Strategic Studies* 39(5–6): 728–61. doi:10.1080/01402390.2016.1208612.	33 pages

October 26: Tech Competition

Activity:	Diffusion Theory (S-curves) Creative insecurity Disruptive Innovation Technology Acceptance Model Adoption-Capacity Theory Offsetting vs. leapfrogging

October 28: Spillover

Lecture:	History of industrial espionage IDAR Model Tacit knowledge Absorption capacity First & Second "Chip Wars"

Week 11: Countermeasures

Required Reading	Length
NSDD 189 (Sep 21, 1985). https://irp.fas.org/offdocs/nsdd/nsdd-189.htm	~1 page
NSPM-33 (Jan 14, 2021). https://trumpwhitehouse.archives.gov/presidential-actions/presidential-memorandum-united-states-government-supported-research-development-national-security-policy/	~4 pages
Walker-Munro, Brendan. 2024. "National Security, Foreign Investment & Research Security: The Current State of Art." *Griffith Law Review* 33(2): 167–88. doi:10.1080/10383441.2025.2459007.	21 pages
Gallagher, Mike. 2025. "Opinion | Send Harvard's Chinese Students Home." *The Wall Street Journal*. https://www.wsj.com/opinion/send-harvards-chinese-students-home-national-security-risk-research-2e43313d (August 21, 2025).	2 pages
Xie, Yu, Xihong Lin, Ju Li, Qian He, and Junming Huang. 2023. "Caught in the Crossfire: Fears of Chinese–American Scientists." *Proceedings of the National Academy of Sciences* 120(27): e2216248120. doi:10.1073/pnas.2216248120.	5 pages
Reif, L. Rafael. 2025. "America's Coming Brain Drain." *Foreign Affairs*. https://www.foreignaffairs.com/united-states/trump-universities-war-america-coming-brain-drain (August 14, 2025).	5 pages
Kuo, Kaiser Y. "The War for Chinese Talent in America, with David Zweig." https://www.sinicapodcast.com/p/the-war-for-chinese-talent-in-america (August 14, 2025).	52 mins
"BBC Audio | The Documentary Podcast | China's Rocket Man." https://www.bbc.co.uk/programmes/w3ct1642 (July 21, 2025).	19 mins

November 2: Knowledge Security

Lecture:	Spillover (continued) Research Security Policies Aptitude vs. “affinity” Brain drain

November 4: Guest Speaker

Lecture:	Aaron Melville. CMU Assistant VP, Research Security.

Week 12: “Boomers” & “Doomers”

Required Reading	**Length**
Maynard, Andrew. 2016. “Why I’m Suffering From Nanotechnology Fatigue.” *Slate*. https://slate.com/technology/2016/09/why-im-suffering-from-nanotechnology-fatigue.html **(August 22, 2025).**	~5 pages
Edwards, Chris. 2015. “Waiting for the Drop.” *Engineering & Technology* 10(4): 32–53. https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=7118566&casa_token=21o65bU1uuQAAAAA:n9mthYLvJB8walkv4shLKVl00xpT47-UCaLsw33TcOGm94CamX6f2fJ9-LGU-3NWhTwxq3U2	21 pages
Kestenbaum, David. 2021. “When Luddites Attack.” *NPR*. https://www.npr.org/2021/09/24/1040606747/when-luddites-attack-classic	19 mins
Correll, John T. “The Reformers.” Air & Space Forces. https://www.airandspaceforces.com/article/0208reformers/	~5 pages

November 9: Inertia

Activity:	Hype cycles Luddites *The Pentagon Wars* (1998) Peacetime vs. wartime Secrecy vs. disclosure Whiz-Kids vs. Reformers Public-private trust Primes vs. Silicon Valley

November 11: Simulation Intro

Activity:	*Pick simulation teams* Simulation instructions

Week 13: Simulation

Required Reading	**Length**
Guo, Jeff. 2024. "The Strange Way the World's Fastest Microchips Are Made." *NPR*. https://www.npr.org/2024/11/13/1212604208/asml-euv-extreme-ultraviolet-lithography-microchips (August 14, 2025).	27 mins
Nelson, Richard. 1962. "The Link Between Science and Invention: The Case of the Transistor." In *The Rate and Direction of Inventive Activity: Economic and Social Factors*, Princeton University Press, 549–84. https://www.nber.org/books-and-chapters/rate-and-direction-inventive-activity-economic-and-social-factors/link-between-science-and-invention-case-transistor (August 14, 2025).	29 pages (p. 554-583)
***The Most Amazing — and Dangerous — Technology in the World*. 2023. https://www.youtube.com/watch?v=kGuL3UQte44 (August 15, 2025).**	58 mins

November 12: Simulation Day 1

Activity:	Strategies & Gameplay

November 18: Simulation Day 2

Lecture:	Recap & Gameplay

Week 14: Clocking the Race

Required Reading	Length
Hurst, Daniel. 2023. "China Leading US in Technology Race in All but a Few Fields, Thinktank Finds." ***The Guardian*****.** **https://www.theguardian.com/world/2023/mar/02/china-leading-us-in-technology-race-in-all-but-a-few-fields-thinktank-finds**	~2 pages
Ding, Jeffrey. 2024. "The Diffusion Deficit in Scientific and Technological Power: Re-Assessing China's Rise." ***Review of International Political Economy*** **31(1): 173–98. doi:10.1080/09692290.2023.2173633.**	25 pages
Wang, Raymond. "Guns and Butter: Measuring Spillover and Implications for Technological Competition." ***Journal of Strategic Studies*** **0(0): 1–31. doi:10.1080/01402390.2024.2445001.**	31 pages (skim)
"How to Measure Innovation | St. Louis Fed." https://www.stlouisfed.org/open-vault/2021/june/how-to-measure-innovation (May 5, 2025).	~1 page
Bearak, Max, and Mira Rojanasakul. 2025. "How China Went From Clean Energy Copycat to Global Innovator." ***The New York Times*****.** **https://www.nytimes.com/interactive/2025/08/14/climate/china-clean-energy-patents.html (August 18, 2025).**	~3 pages
Ang, Yuen Yuen, Nan Jia, Bo Yang, and Kenneth G. Huang. 2024. "China's Low-Productivity Innovation Drive: Evidence From Patents." ***Comparative Political Studies*** **57(12): 2011–45. doi:10.1177/00104140231209960.**	34 pages

November 23: Simulation Day 3

Lecture:	Final round of gameplay Tally points & discuss Diffusion deficits

November 25: THANKSGIVING HOLIDAY

Activity:	No class. Enjoy!

Week 15: Course Conclusion

Required Reading	Length
Pottinger, Matt, and Mike Gallagher. 2024. "No Substitute for Victory." ***Foreign Affairs*** **103(3).** **https://www.foreignaffairs.com/united-states/no-substitute-victory-pottinger-gallagher (August 14, 2025).**	~5 pages
Truex, Rory. 2021. "What the Fear of China Is Doing to American Science." ***The Atlantic*****.** **https://www.theatlantic.com/ideas/archive/2021/02/fears-about-china-are-disrupting-american-science/618031/ (August 14, 2025).**	~5 pages
Liu, Cixin. 2018. *Ball Lightning*. New York: Tor Books. ● Fun winter break fiction exploring course themes! Not required. ● A used copy can be purchased for as little as $3 on Amazon. You may choose to read it in either English or the original Chinese.	**(optional)**

November 30: Wrapping Up

Lecture:	Technological revolutions The Era of Open Innovation Is technology speeding up? Sim reports due

December 2: Final Exam

Activity:	In-class exam.

Week 16: Finals Week

Independent reading. Grade deadline: December 16.